

ORAL MUCOSA

LINNING MUCOSA

Lecture learning outcomes

At the end of this lecture the student should be able to :

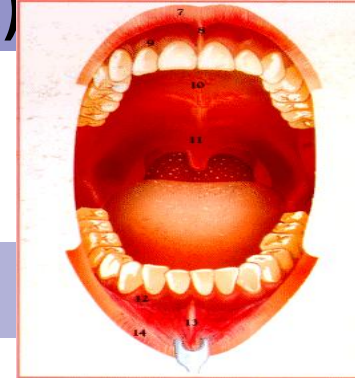
1. Discuss the different types of lining mucosa.
2. Describe microscopic structure of different types of lining mucosa

Classification of oral mucous membrane

1-Keratinized mucosa (Masticatory mucosa)

(A) Gingiva

(B) Hard palate



2- Non-keratinized mucosa (Lining mucosa)

(A) Firmly attached

(B) Loosely attached

**Soft
palate**

lip

cheek

**Ventral S
tongue**

**Floor of
mouth**

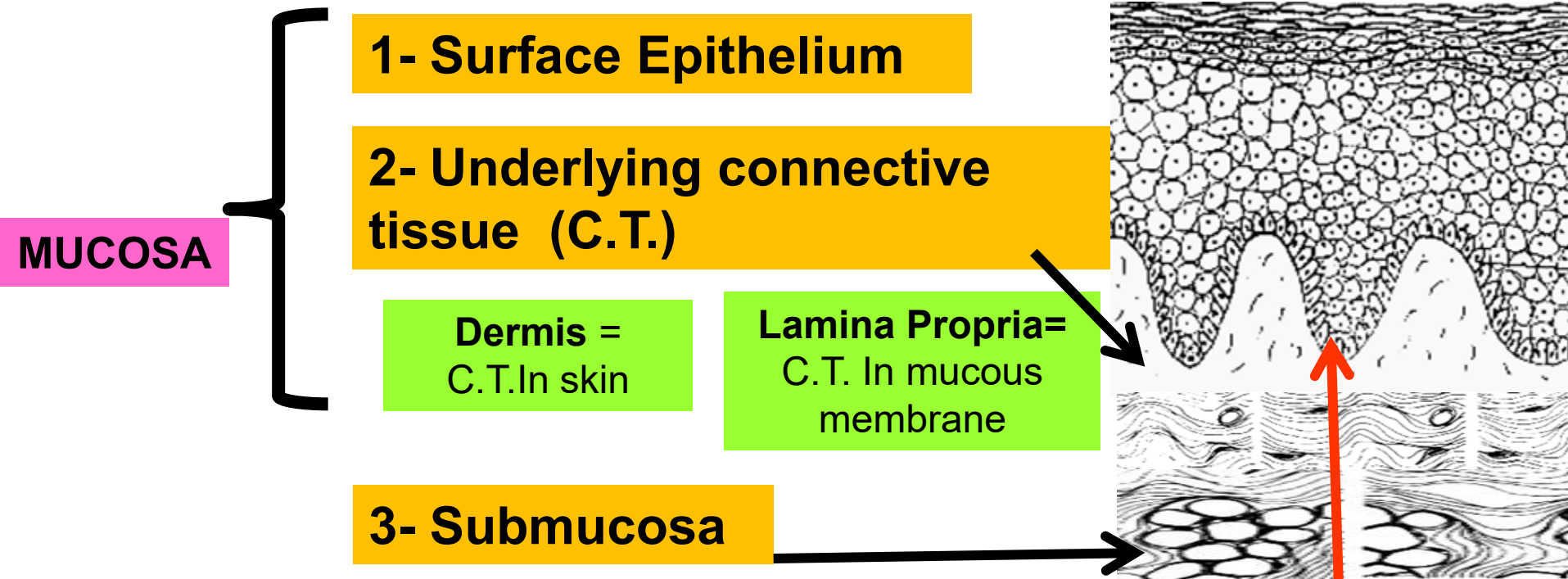
Vestibule

**Alveolar
mucosa**

3- Specialized mucosa

Dorsal surface of the tongue

Structure of mucous membrane



The epithelium is separated from the underlying lamina propria by **BASEMENT MEMBRANE**

Downward projections of the epithelium in the underlying C.T. is called rete-pegs or rete-ridges

Lining mucosa

Firmly attached **(covering ms.)**

- Soft palate
- Inferior surface of the tongue
- Lip
- cheek

Loosely attached **(allow free movement)**

- Alveolar mucosa
- Vestibular fornix
- Floor of the mouth

Loosely Attached "Movable" mucosa

Alveolar mucosa

loosely attached
to periosteum of alveolar bone

Vestibular fornix

allows mobility
of lips and cheeks

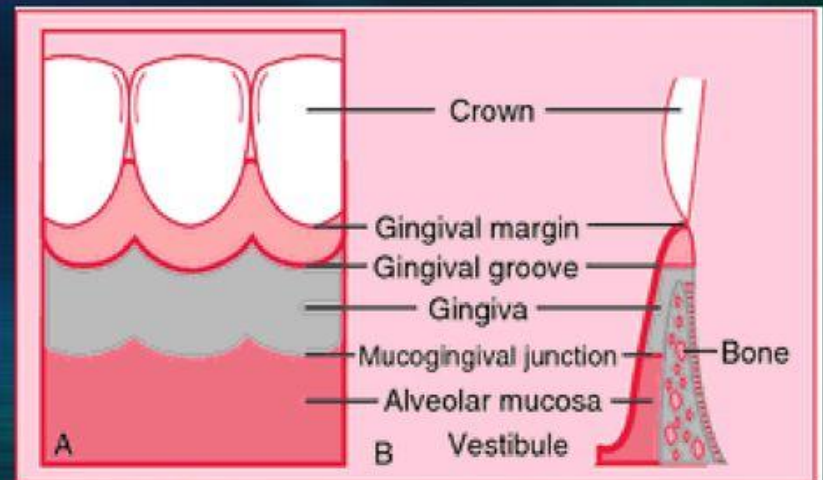
Floor of the mouth

loosly attached to
underlying structures
allows mobility of the tongue



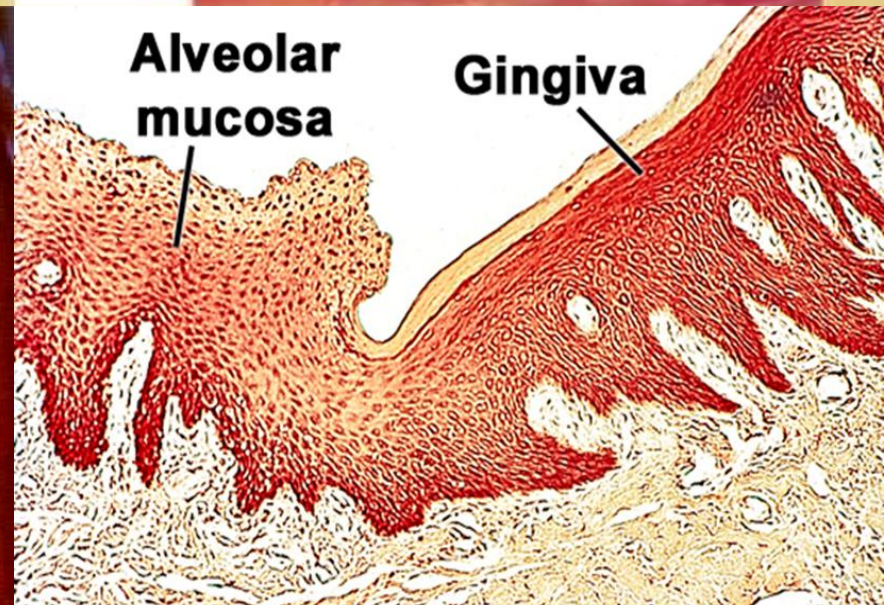
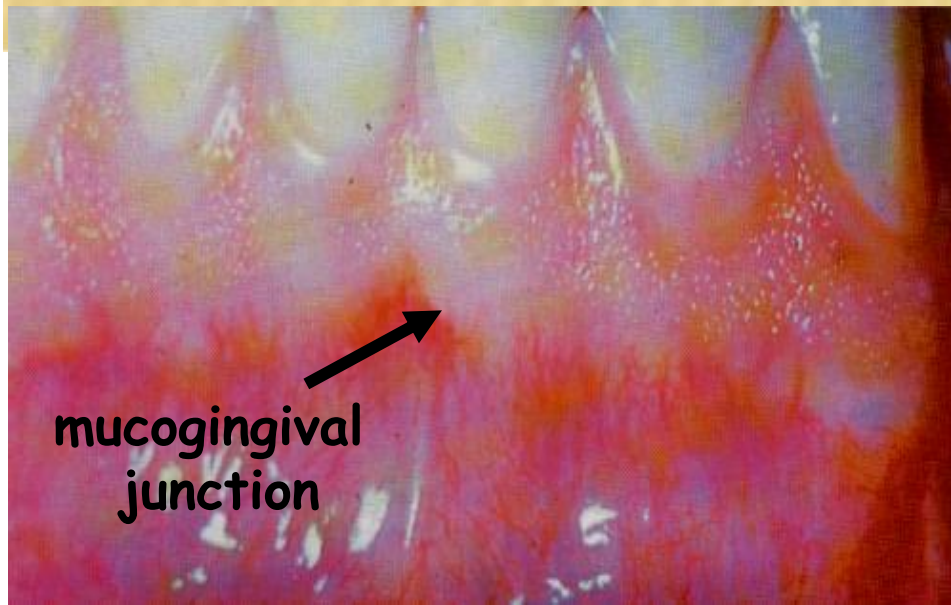
Definitions

- 1 **Alveolar Mucosa** – Loosely attached mucosa covering the basal part of the alveolar process and continuing into the vestibular fornix and the floor of the mouth



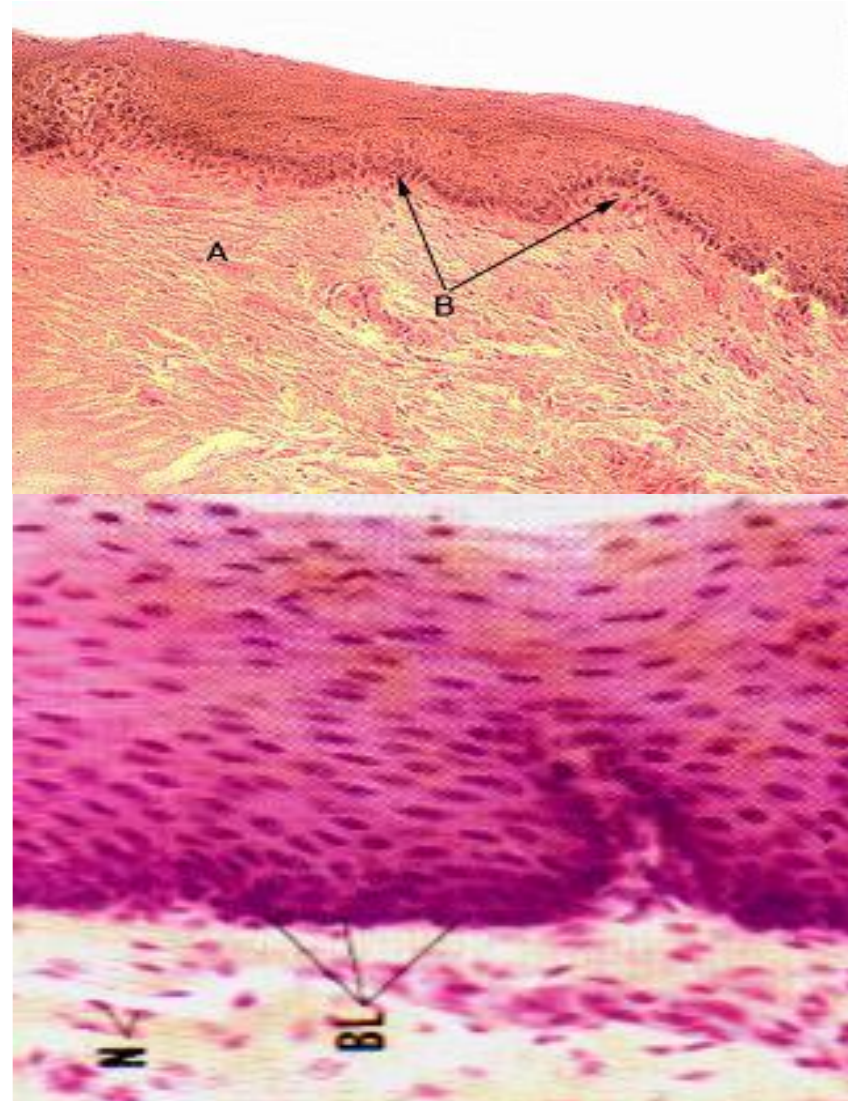
Alveolar Mucosa (Anatomy)

The alveolar mucosa is separated from the periosteum by a loose, highly vascular connective tissue. Thus the alveolar mucosa is relatively loose and mobile tissue, deep red, in marked contrast to the pale pink attached gingiva.



Alveolar Mucosa (Histology)

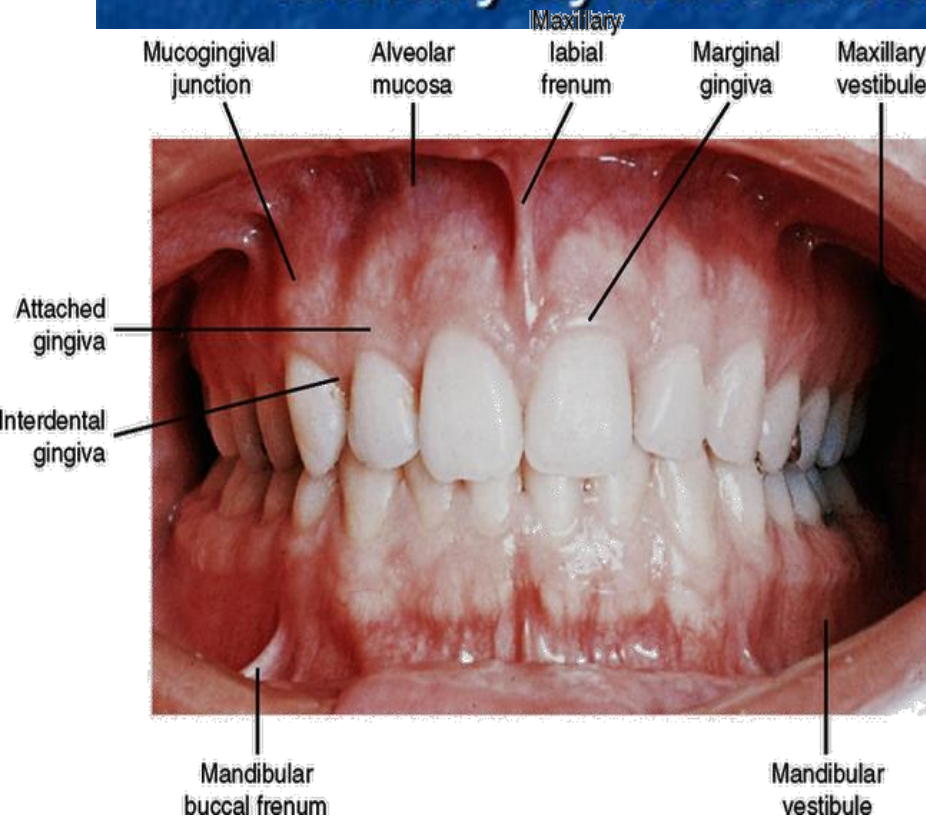
- 1. Epithelium:**
Thin, non-keratinized (with few or missing rete-pegs)
- 2. Lamina Propria:**
 - CT papillae sometimes missing.
 - Collagen fibers, many elastic fibers.
 - Extensive Vascular Supply
- 3. Submucosa:**
Present with minor mixed salivary glands and many thick elastic fibers
- 4. Loosely attached to the underlying alveolar process**



2-Vestibular fornix (Anatomy)

It is bound facially by mucosa of lips , cheek & orally by **Alveolar mucosa** .

Vestibule is divided medially by labial frenum & laterally by buccal frenum.



- The mucosa of the lips and cheeks reflects from the vestibular fornix to the alveolar mucosa covering the bone.

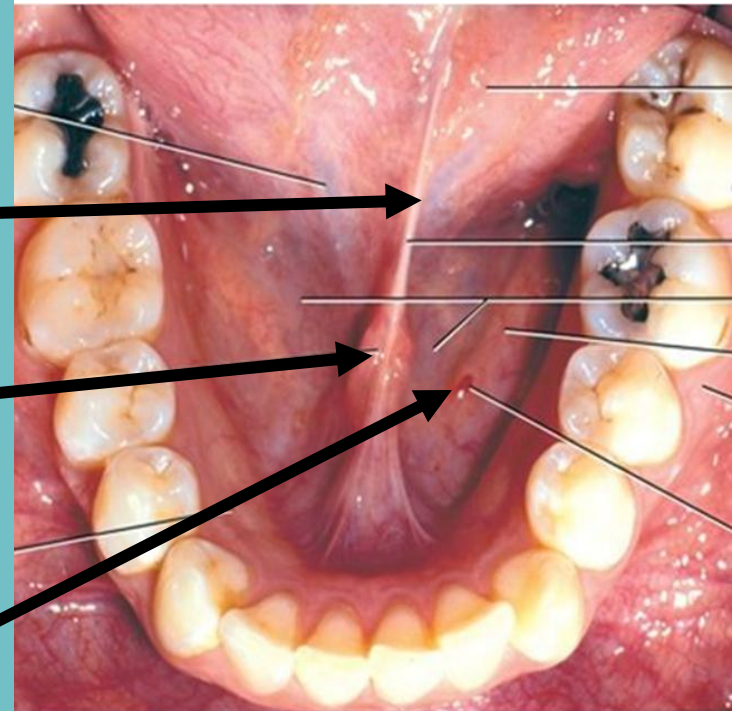
Vestibular fornix (Histology)

Epithelium is thin & nonkeratinized.
Submucosa is thick ,
Large amount of loose areolar tissue.



3-FLOOR OF THE MOUTH

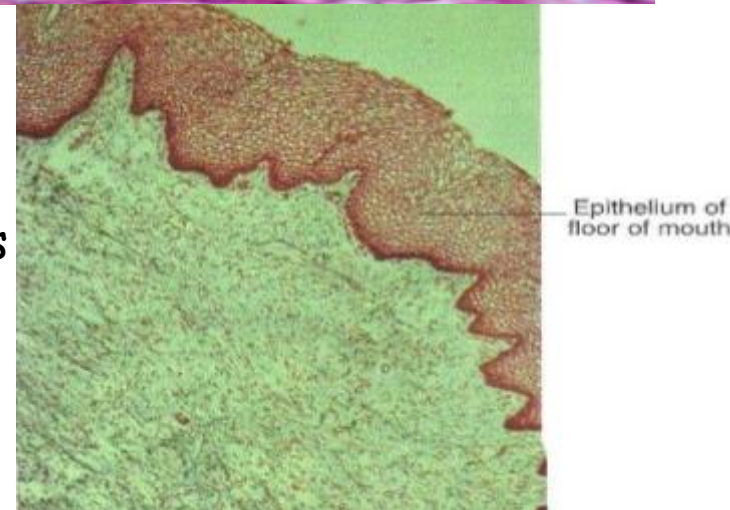
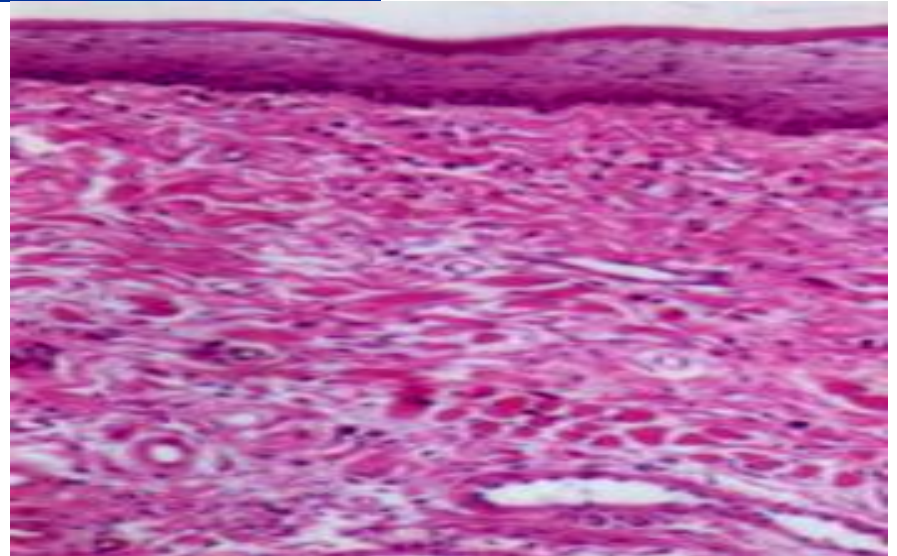
- Formed of mylohyoid & geniohyoid muscles, Covered with mucous membrane
- In the mid line, a mucosal fold (LINGUAL FRENUM) connects the tongue with the floor of the mouth and allow the anterior part to move
- On each side of the frenum there is a small papilla (sublingual caruncle) that has the opening of the duct of the submandibular gland
- A rounded ridge (Sublingual fold) extending backwards and laterally from the papilla, which is produced by the sublingual gland and contain the ducts of sublingual gland



Floor of the mouth (Histology)

- **Epithelium:** Very thin epithelium, non keratinized
- **Lamina propria:**
 - highly vascularized
 - Some elastic fibers
 - Short CT papillae
- **Submucosa:** Present
 - contain adipose tissue with minor salivary Gs

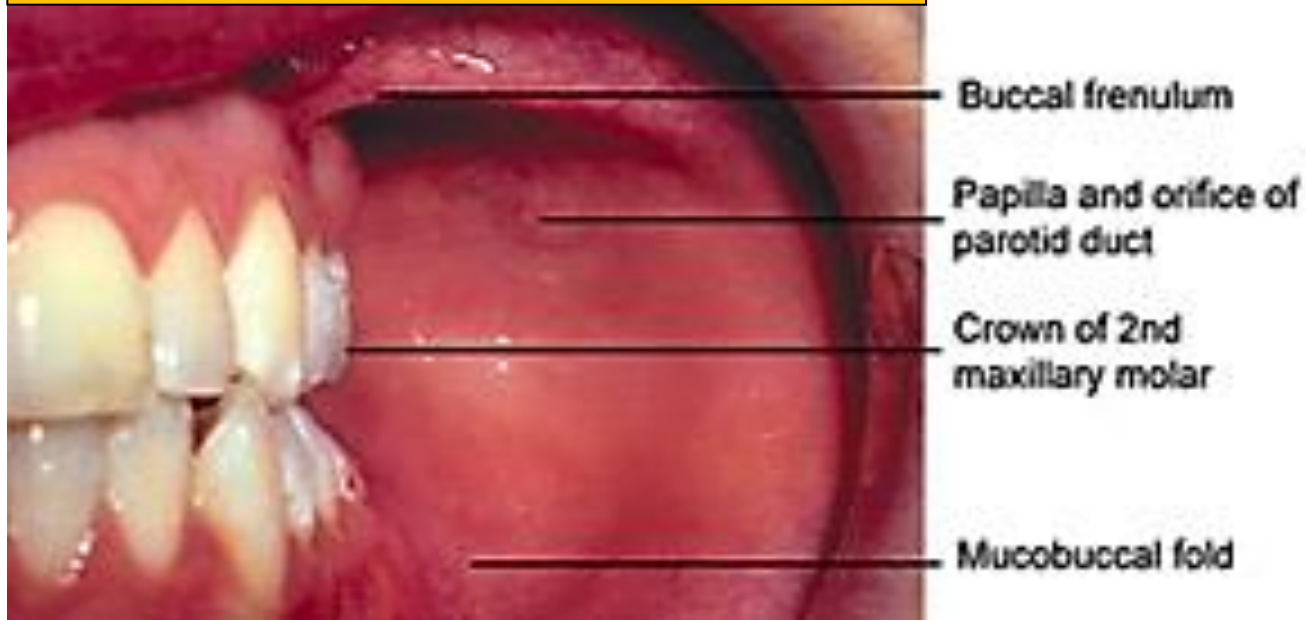
Loosely attached to the underlying structures
sublingual and submandibular SGs



**FIRMLY ATTACHED
LINNING MUCOSA**

1- Buccal Mucosa

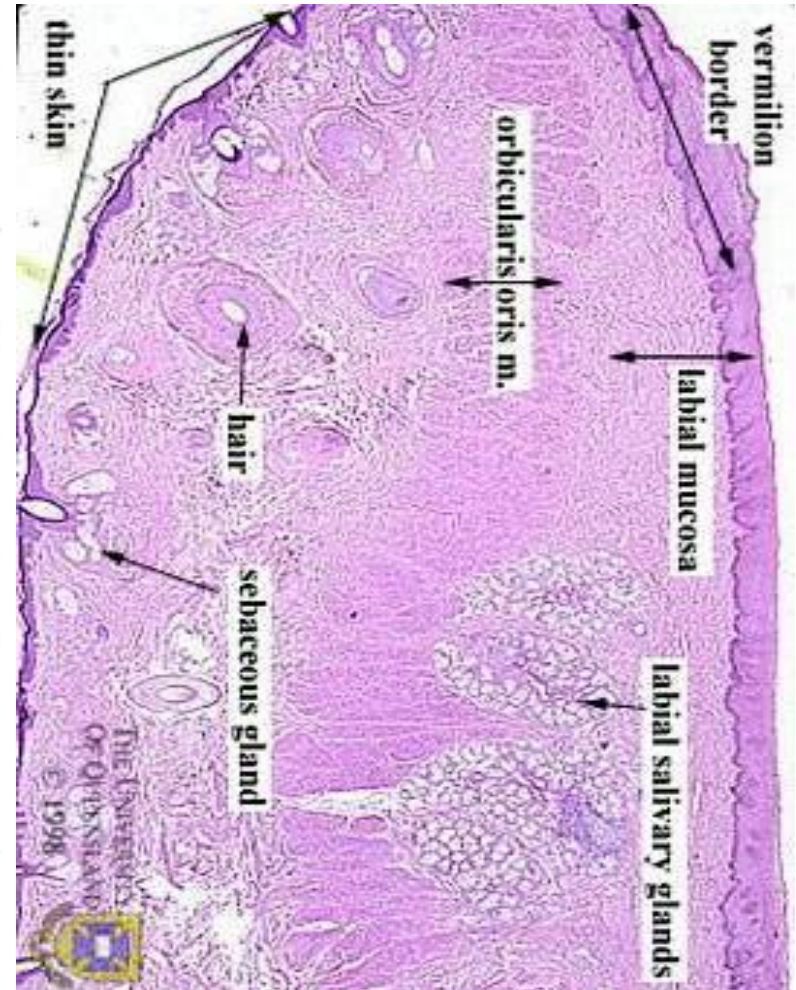
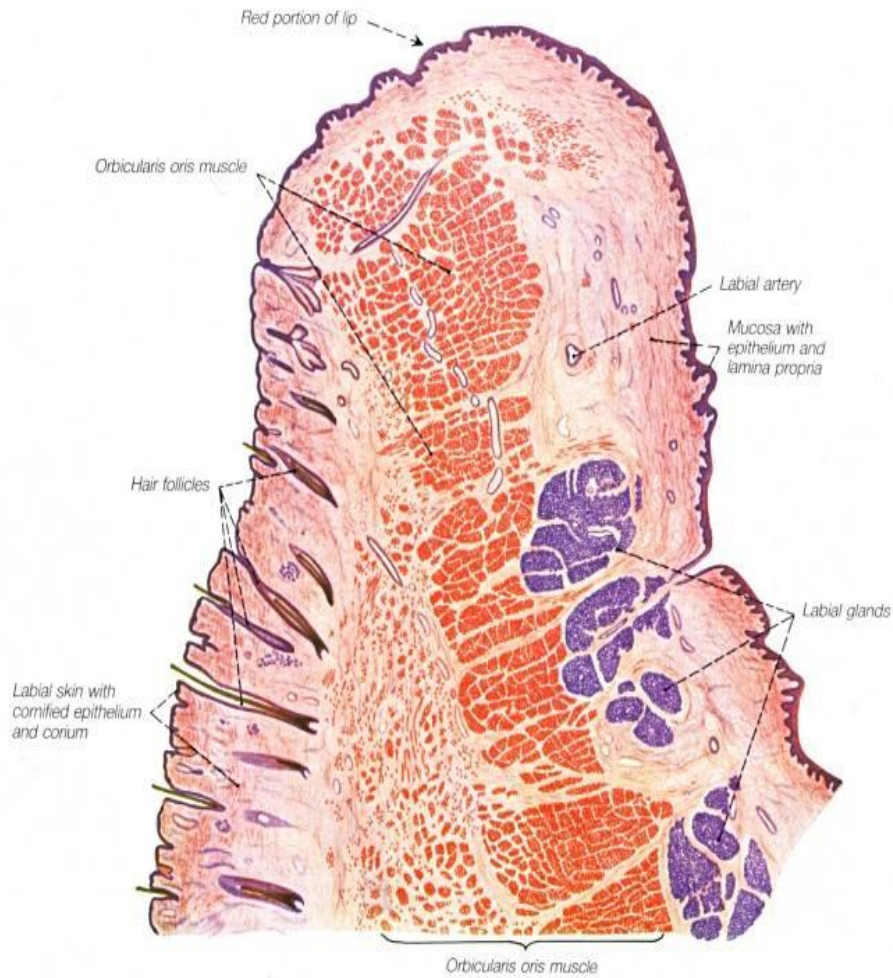
(Anatomy)



The buccal [mucosa](#) is defined as; the epithelium lining the inner surface of the cheeks

- The buccal [mucosa](#) is comprised of the [mucosal surfaces](#) of the cheeks.
- Stensen's (parotid) ducts are located within the buccal mucosa at the parotid papillae corresponding to the [second maxillary molars](#).

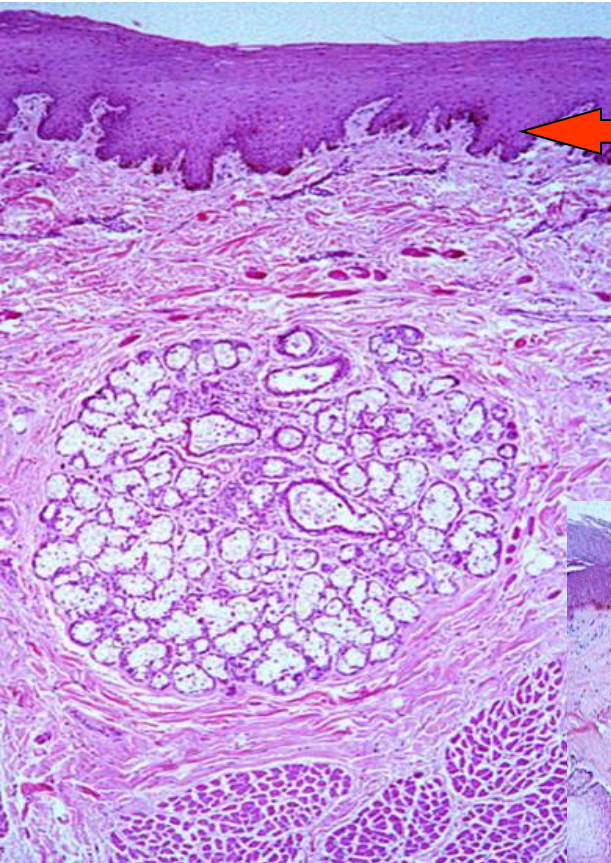
Lip



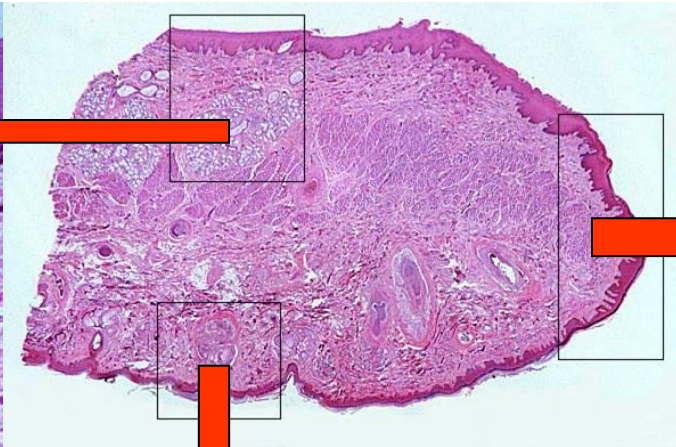
Lip

**2-Mucous side
(Labial mucosa)**

**Vermilion border
(Mucocutaneous
Junction)**



**Submucosa contain
fat cells & minor
mixed S.Gs**

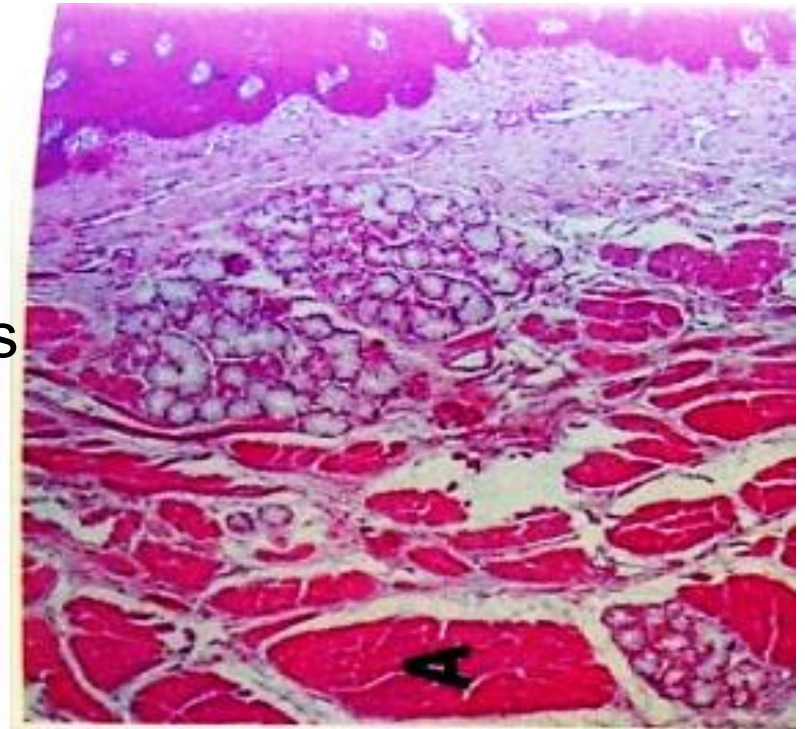


Skin side

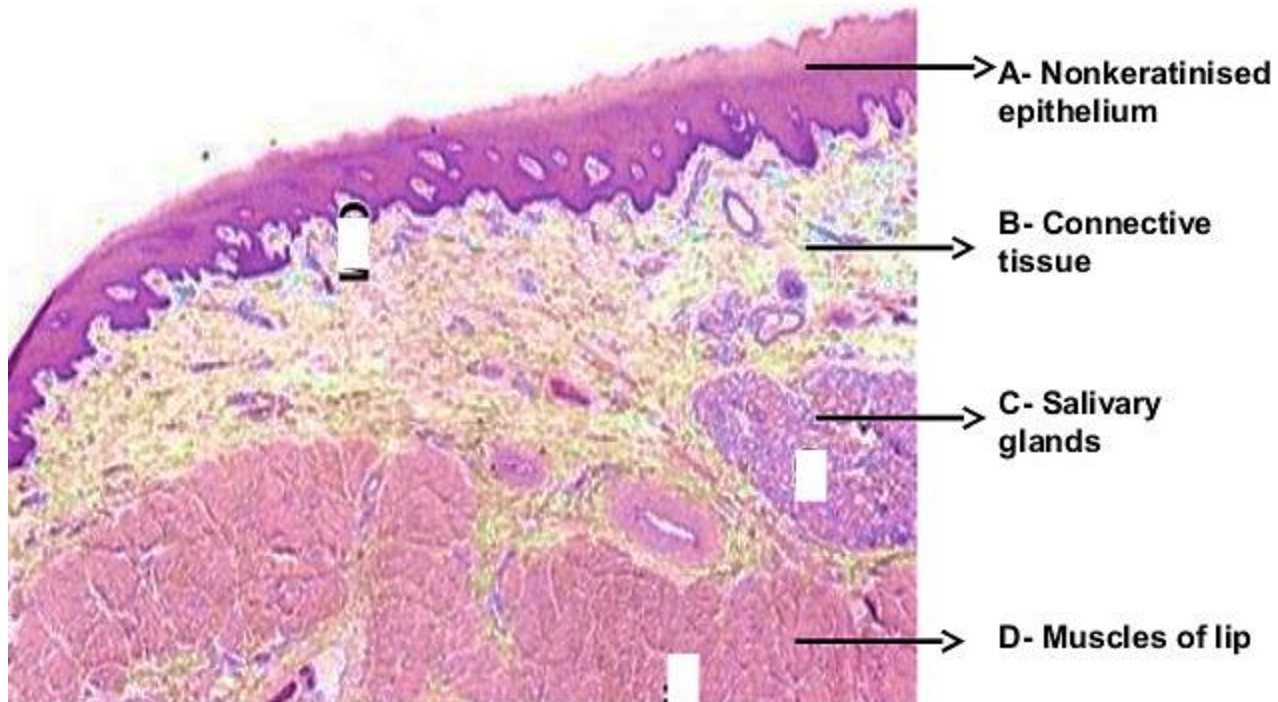
LABIAL AND BUCCAL MUCOSA

- **Epithelium:** Thick st. sq. non-keratinized epith.
- **Lamina Propria:** - Irregular & blunt CT papillae
 - Some elastic fibers
 - Extensive vasculature
- **Submucosa:** - Present
 - Adipose tissue
 - Minor salivary glands

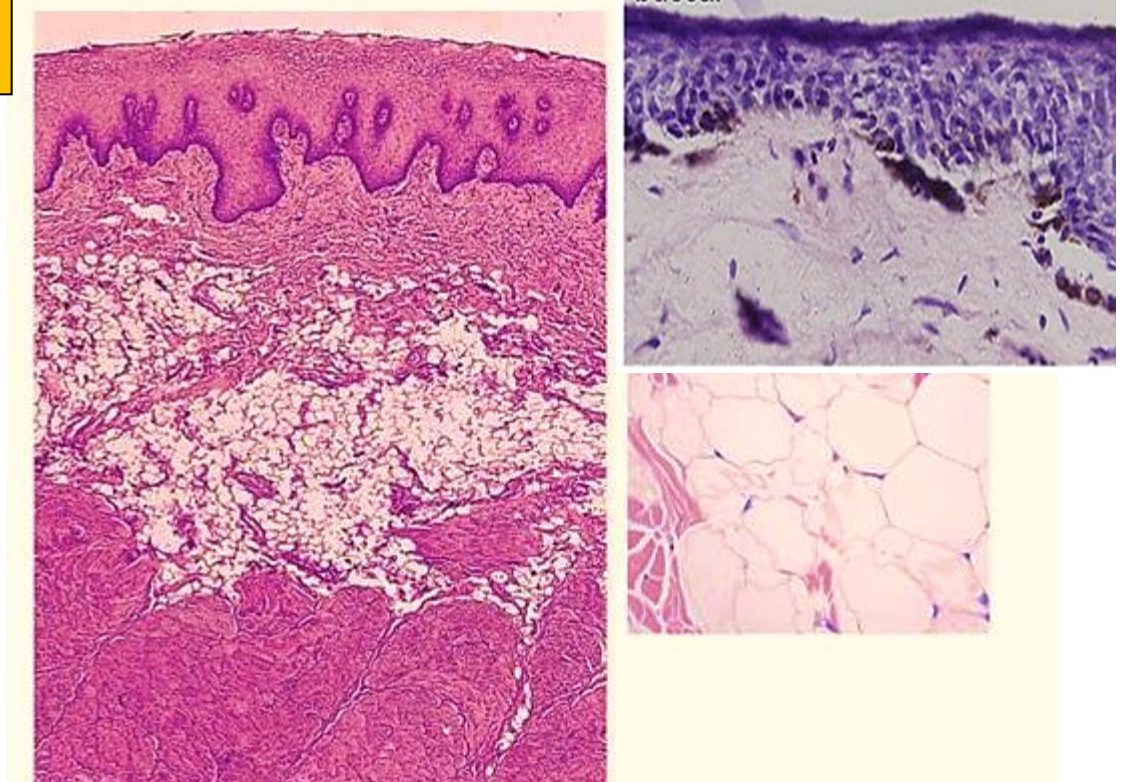
Firmly attached to the underlying muscle.



HISTOLOGY OF LABIAL MUCOSA



Buccal mucosa

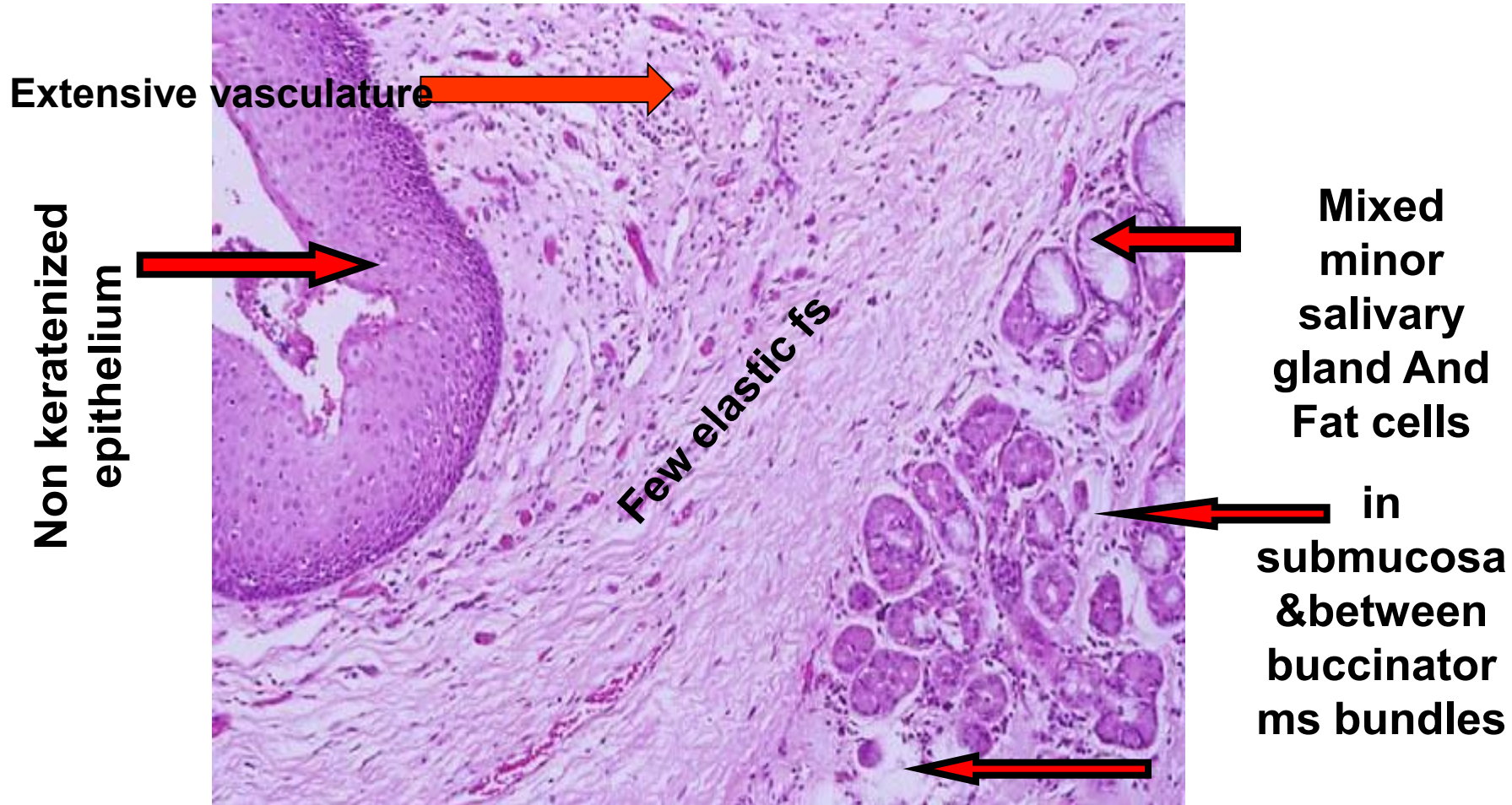


Nonkeratinized stratified squamous epithelium, lamina propria and submucosa

Submucosa of cheeks contain fat cells along with lobules of minor salivary glands and muscle fibers

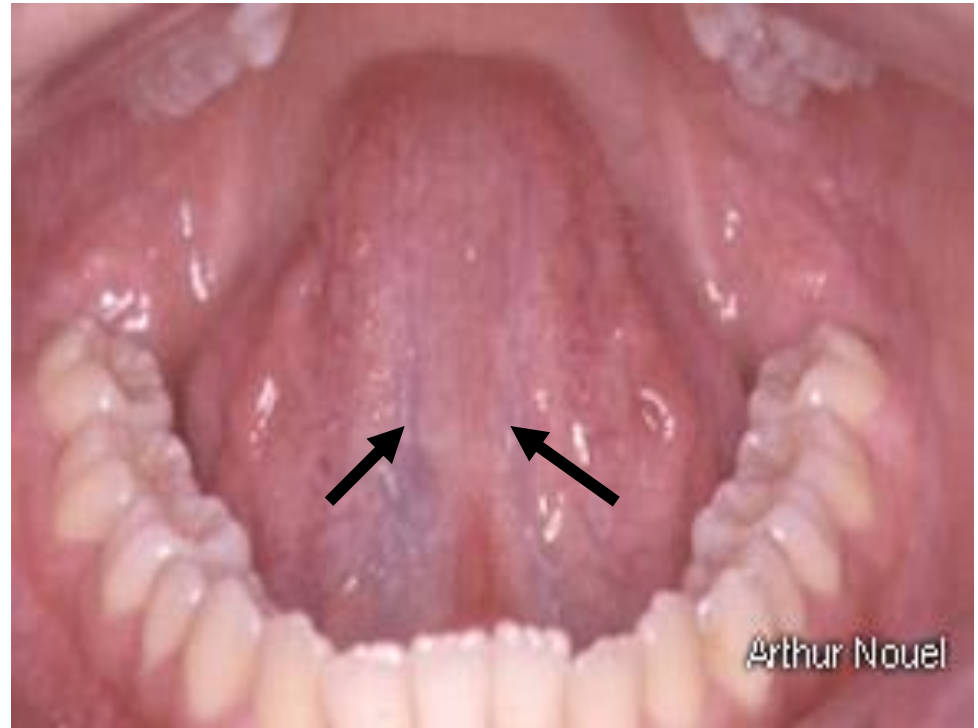
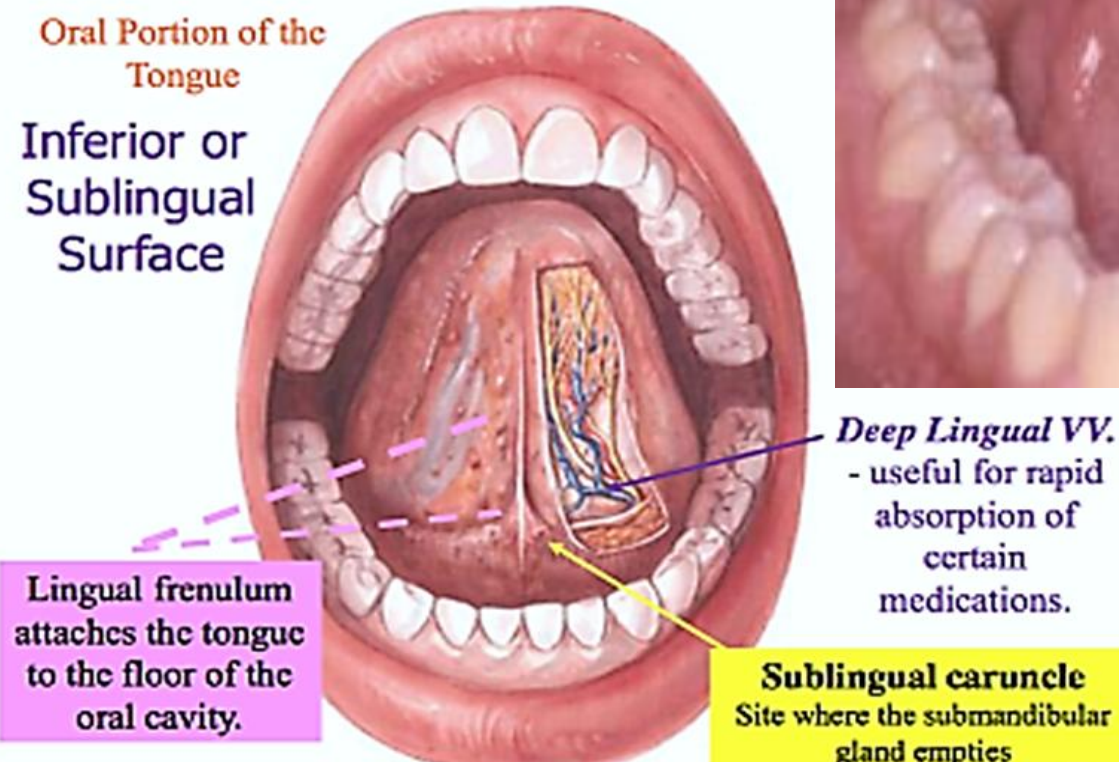
- N.B.: Elastic fibers found in lamina propria prevent folding of the mucosa & mixed minor salivary glands for lubrication of mouth

Cheek(buccal) mucosa

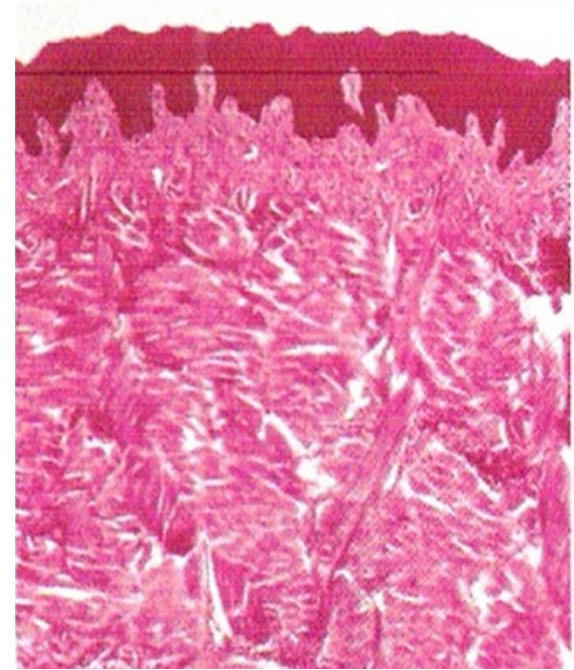
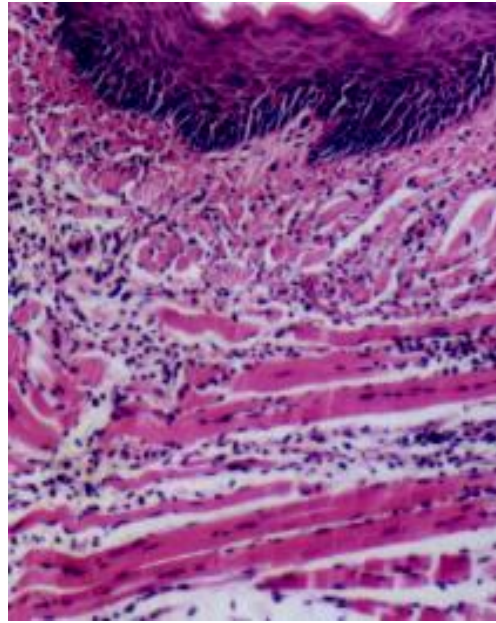
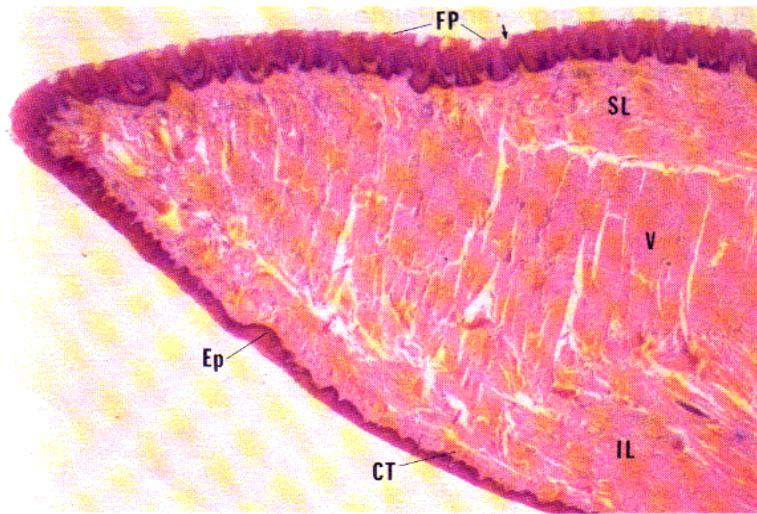


3-Inferior surface of the tongue (ANATOMY)

- On each side of the frenulum, a **deep lingual vein** is visible through the thin mucous membrane.



Ventral (inferior) surface of the tongue

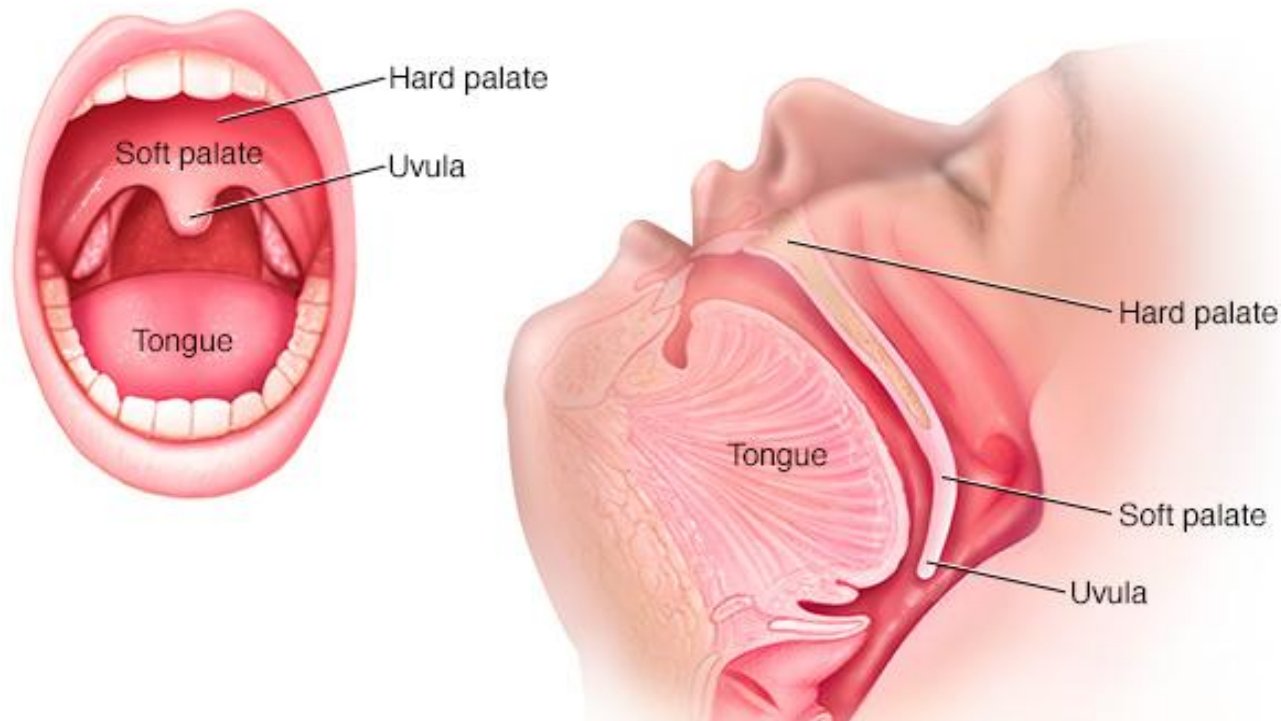


HISTOLOGY:

- Firmly attached to underlying muscles
 - Thin non.kert. st.sq. epith.
- Saw-teeth appearance** of epith ridges. CT Papillae are numerous but short
- No distinct layer of submucosa as it merges with the muscle fibers of the ventral surface of the tongue

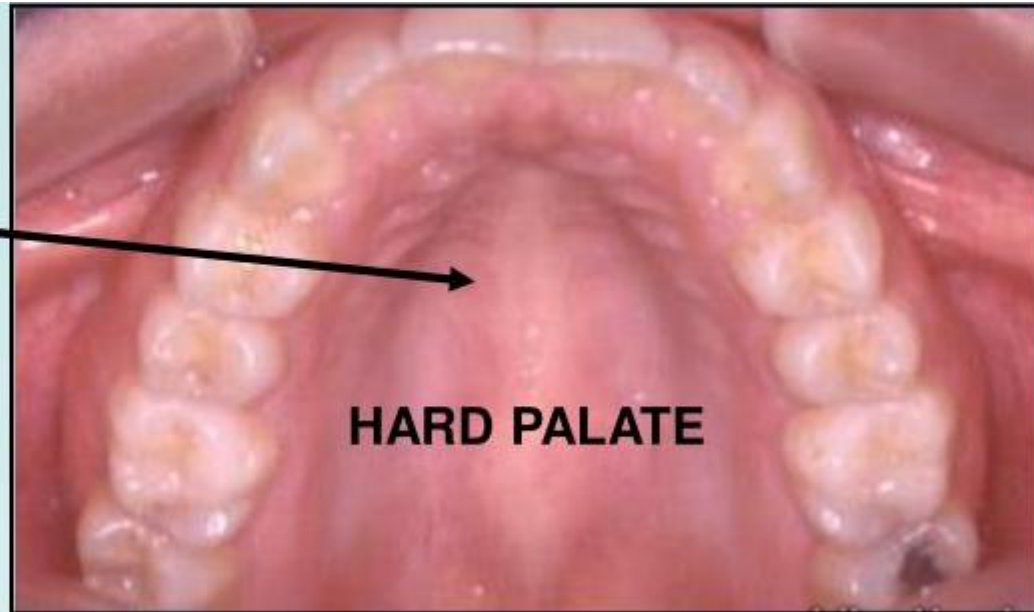
4- SOFT PALATE

- Soft Palate is the soft tissue constituting the back of the roof of the mouth. .
- Unlike the hard palate, the soft palate doesn't contain a bony framework.

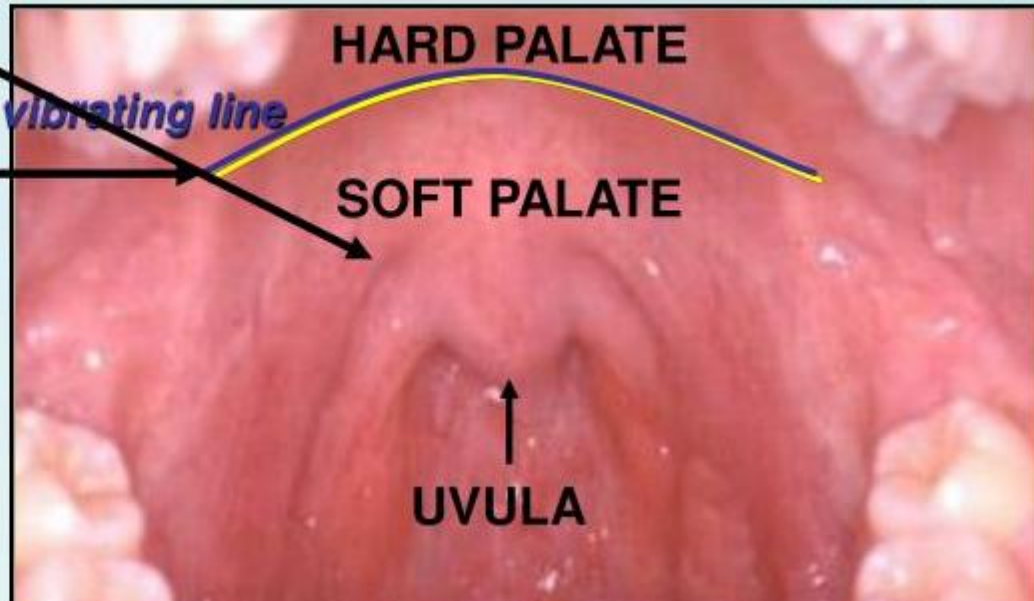


PALATE

- Its **anterior part** is named as **hard palate**.
- Its **posterior part** is named as **soft palate** which ends by a small fleshy structure named **uvula** hanging from its center.
- The **junction** between hard and soft palate is called **vibrating line**.
- The **soft palate** is sometimes **redder** than the hard palate.



HARD PALATE



HARD PALATE

SOFT PALATE

UVULA

SOFT PALATE:

1- Epithelium:

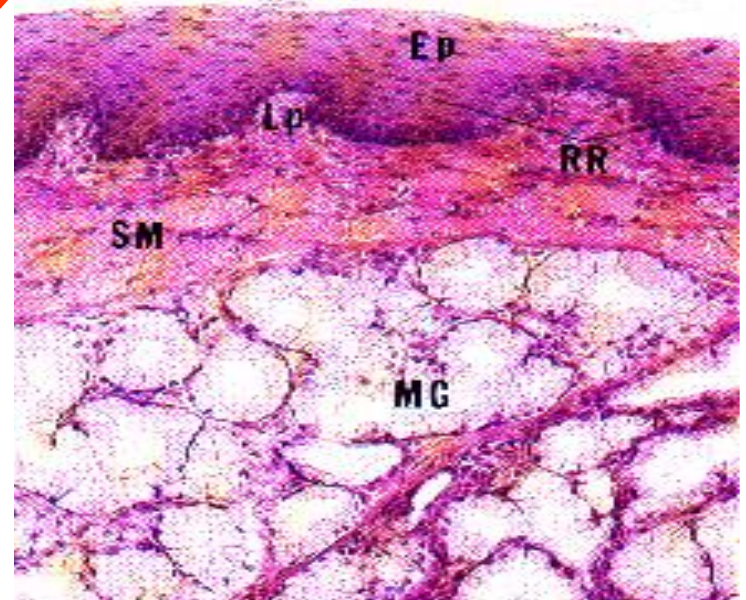
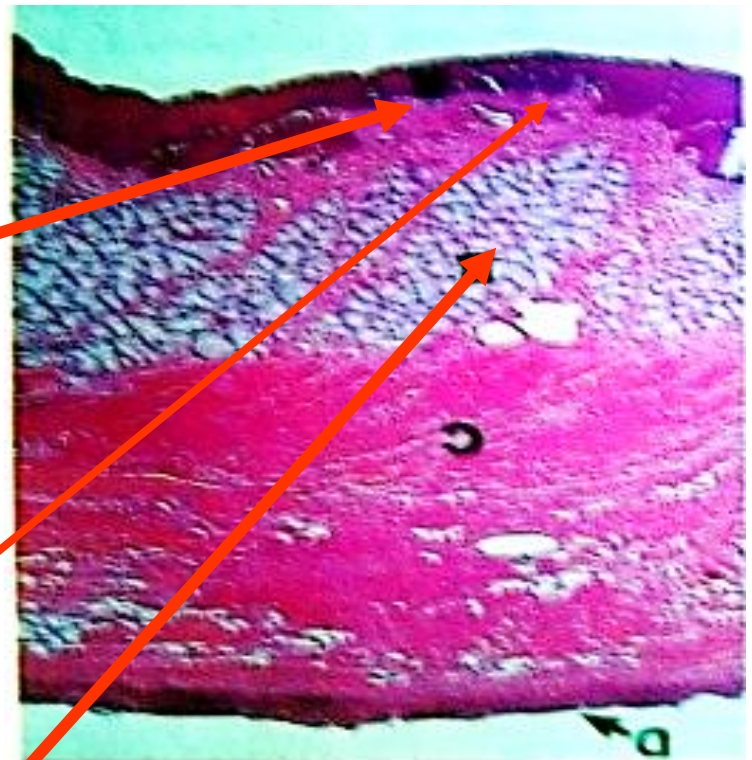
Thin non-keratinized

2- Lamina propria:

Thick with numerous CT papillae and distinct ELASTIC LAYER, Highly vascular

3- Submucosa : very thin with Adipose tissue & minor salivary glands

Firmly attached to underlying muscles.





THANK YOU